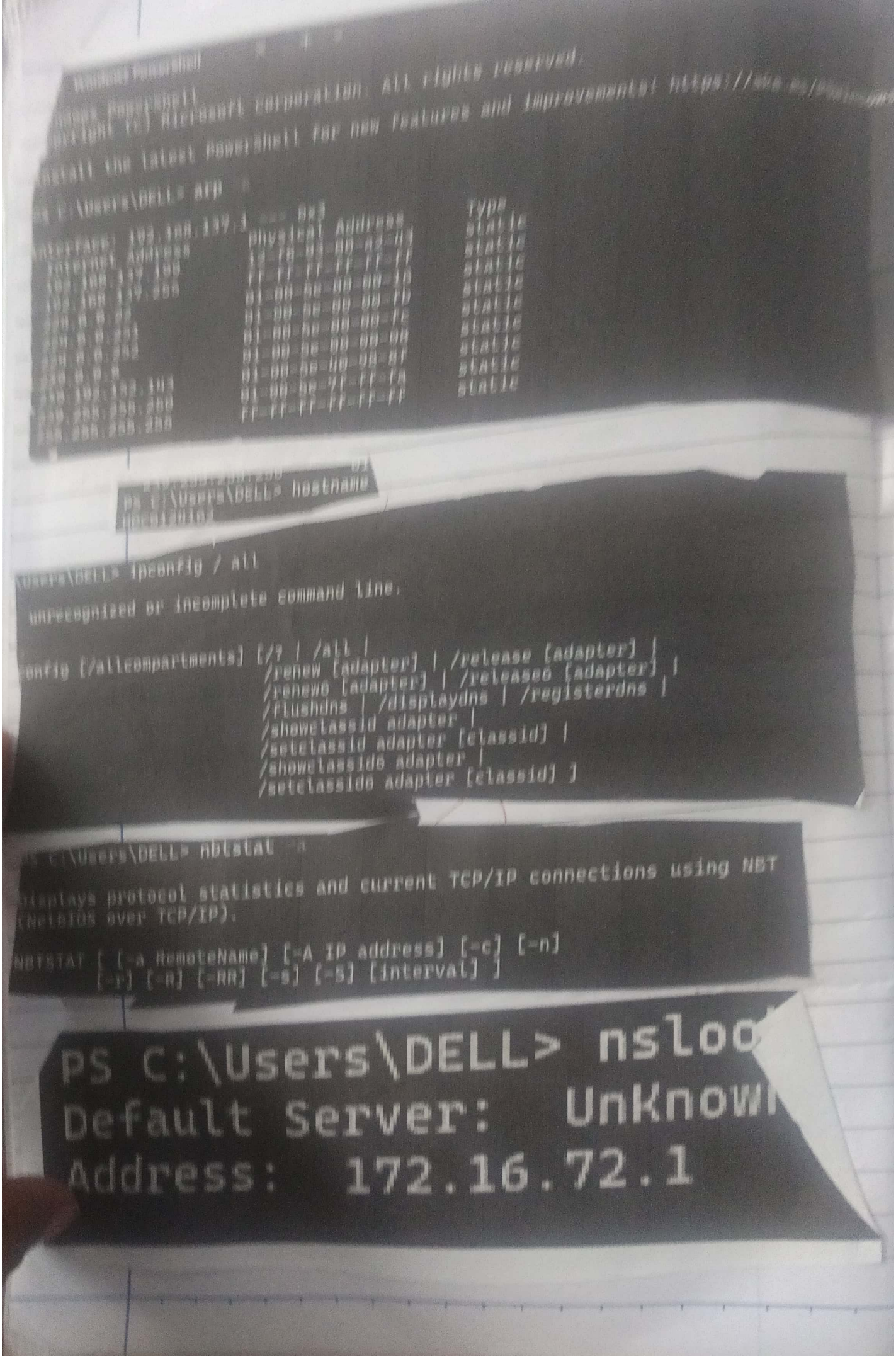

Physical Address
Type
static
PS C:\Users\DELL> hostname
ipconfig / all
Unrecognized or incomplete command line.
ipconfig [/allcompartments] [/? | /all |
/renew [adapter] | /release [adapter] |
/renew6 [adapter] | /release6 [adapter] |
/flushdns | /displaydns | /registerdns |
/showclassid adapter |
/setclassid adapter [classid] |
/showclassid6 adapter |
/setclassid6 adapter [classid]]
PS C:\Users\DELL> nbtstat -a
Displays protocol statistics and current TCP/IP connections using NBT
(NetBIOS over TCP/IP).
NBTSTAT [[-a RemoteName] [-A IP address] [-c] [-n]
[-r] [-R] [-RR] [-s] [-S] [interval]]
PS C:\Users\DELL> nsloo
Default Server: UnKnow
Address: 172.16.72.1

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6/7/26 Basic Network Commands

Aim:-
Basic Networking Commands in Linux and windows operating System

Basic Networking Commands:-

(1) arp -a :- ARP is short form of address resolution Protocol and find address of your router.

(2) Hostname:- This is the simplest of all Tcp/Ip commands display the computer o/p.

(3) ipconfig /all:- This Commands displays detail configuration information all connection including router, Internet and Type of System.

(4) nbtstat -a:- network Statics is display a Variety of Statics about a Computer Tcp/Ip connection.

(5) nbstat -a :- This commands helps solve Problem with NetBios name resolution.

(6) nslookup:- name Server lookup is a tool used to Perform Dns lookup and display Dns details.
Eg:- nslookup www.google.com.

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```
C:\Users\DELL>pathping

Usage: pathping [-g host-list] [-h maximum_hops] [-i address]
                [-p period] [-q num_queries] [-w timeout]
                [-4] [-6] target_name

Options:
   -g host-list       Loose source route along host-list.
   -h maximum_hops    Maximum number of hops to search for
   -i address         Use the specified source address.
   -n                 Do not resolve addresses to hostnames
   -p period          Wait period milliseconds between ping
   -q num_queries     Number of queries per hop.
   -w timeout         Wait timeout milliseconds for each re
   -4                 Force using IPv4.
   -6                 Force using IPv6.
```

```
PS C:\Users\DELL> ping

Usage: ping [-t] [-a] [-n count] [-l size] [-f] [-i TTL] [-v TOS]
            [-r count] [-s count] [[-j host-list] | [-k host-list]]
            [-w timeout] [-R] [-S srcaddr] [-c compartment] [-p]
            [-4] [-6] target_name
```

Windows PowerShell × Windows PowerShell

```
PS C:\Users\DELL> route

Manipulates network routing tables.

ROUTE [-f] [-p] [-4|-6] command [destination]
                  [MASK netmask]  [gateway] [METRIC metric]  [IF i

  -f           Clears the routing tables of all gateway entries.
               used in conjunction with one of the commands, the ta
               cleared prior to running the command.

  -p           When used with the ADD command, makes a route persist
               boots of the system. By default, routes are not prese
               when the system is restarted. Ignored for all other c
               which always affect the appropriate persistent routes.
```

MATE Terminal

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```
                                        My traceroute  [v0.95]
fedora (: 1) -> localhost (: 1)
Keys:  Help   Display mode   Restart statistics
                                             Packets
Host                                       Loss%   Snt
1. localhost                                0.0%    30
```

MATE Terminal

File Edit View Search Terminal Help

```
$ ip
Usage: ip [ OPTIONS ] OBJECT { COMMAND | help }
       ip [ -force ] -batch filename
where  OBJECT := { address | addrlabel | ant | fou | help | ila | ioam | l2tp |
                   link | macsec | maddress | monitor | mptcp | mroute | mrule |
                   neighbor | neighbour | netconf | netns | nexthop | ntable |
                   ntbl | route | rule | sr | tap | tcpmetrics |
                   token | tunnel | tuntap | vrf | xfrm }
       OPTIONS := { -V[ersion] | -s[tatistics] | -d[etails] | -r[esolve] |
                    -h[uman-readable] | -iec | -j[son] | -p[retty] |
                    -f[amily] { inet | inet6 | mpls | bridge | link } |
                    -4 | -6 | -M | -B | -0 |
                    -l[oops] { maximum-addr-flush-attempts } | -br[ief] |
                    -o[neline] | -t[imestamp] | -ts[hort] | -b[atch] [filename]
```

MATE Terminal

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```
                -rc[vbuf] [size] | -n[etns] name | -N[umeric] | -a[l
                -c[olor]]
              $ ifconfig
lo: flags=73<UP,LOOPBACK,RUNNING,MULTICAST>  mtu 1500
        inet 10.0.2.15  netmask 255.255.255.0  broadcast 10.0.2.255
        inet6 fe80::25f8:2771:cb55:7bf3  prefixlen 64  scopeid 0x20<lin
        ether 08:00:27:9b:47:4d:f4  txqueuelen 1000  (Ethernet)
```

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7) Pathping :- Pathping is unique to Windows and typically a combination of Internet Commands.

8) Ping :- (Packet Internet Groper) Command is the best Ping use Icmp.

9) Route :- Route command is used to view / manipulate the Ip routing

Linux Networking Commands :-

1) Ip :- Displays and configures Ip addresses & network interfaces and routing
2) ifconfig :- Configures and displays network interface settings.
3) mtr :- Combines Ping and traceroute to diagnose network paths and packet loss
4) tcpdump :- Capture and analyze network packets for Troubleshooting
5) Ping :- Test connectivity between two hosts using Icmp Echo request / Reply Messages.